9.8 Driving devices

Driving devices consist of different types of motors and gear trains. They convert the instructions from MCU, into accurate mechanical displacements of the machine tool slides.

Servomotor, stepper motor, synchros and resolver are different types of motors used as drives in NC machines.

9.9 Machine tool

Machine tool is the machine which converts the raw materials into finished components. It is designed to perform various machining operations.

It consists of a machine table, spindles, cutting tools, work holding devices such as jigs and fixtures, coolant systems, swarf removal systems and other auxiliary equipments.

9.10 Classification of CNC Machine tools:

CNC machine tools are classified as follows



1. According to the type of power to the drives
   a. Electrical
   b. Hydraulic
   c. Pneumatic
2. According to motion control system of slides
   a. Point-to-point System
   b. Continuous path system
3. According to the feedback system
   a. Open loop system
   b. Closed loop system
4. According to number of axis
   a. 2 – axis b. 3 – axis
   c. 4 – axis d. 5 – axis

Motion control unit

The cutting tool and the work are located at certain positions in NC machine. During the machining, they are moved from their positions with relation to each other. This system is known as "motion control system"

Point to Point motion control system

Machining is performed only after the cutting tool and the works are located (positioned) at defined positions and there will not be any change in their positions during machining. This system is also called positioning system for this reason.

The relocation is achieved by their movement in two main axes (x axis and y axis). This is the reason for this system to be called as point-to-point system. This system is used in drilling machines, jig boring machines and spot welding machines.

Continuous path motion control system

In this system, both the cutting tool and the work change their positions depends all axes during machining. This system is also called as 'contouring system' and is mainly used in milling machine.

Feed back control system

The actual output movement of NC machines is compared with the desired value as given in the input media. This type of control system is called as "feedback control system".

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